

mercial building consumes over 70 per cent less energy than conventional buildings.

Substantial water savings. Covestro treats 100 per cent wastewater onsite for irrigation and other non-potable purposes. Water utilisation initiatives have enabled it to reduce water usage by up to 50 per cent.

Controlled indoor conditions. Additional setup like optimised use of chiller machines, cooling towers and heat recovery system enables Covestro to maintain a consistently comfortable indoor temperature of 24 degrees celsius at a relative humidity of less than 55 per cent.

The final word

While green constructions such as Covestro's might require a high initial investment, the financial savings in operational costs and positive impact on the environment make them the way to go. **EFY**

—Paromik Chakraborty, technical journalist, EFY

Automated Irrigation: Stepping Stone To Productive Farming

► Optimised conditioning of soil and crops can increase yield by 15-30 per cent

Automation is finding its way to help farmers get the best out of their yards. Let's take a look into how the agricultural community of India can benefit from sensor-driven automated irrigation setups.



Automated irrigation system

Challenges in manual irrigation

Irrigation is the process of applying the right amount of usable water at the right time to cultivation lands and plants to facilitate the yield of a good volume and variety of crops. There are numerous irrigation methods, but in all cases providing the right amount of water at the right time is the common difficult task in manual control.

Under-application of water leads to numerous problems including increased salinity of soil or insufficient moisture content, which might result in wilting of plants, drying of crops, infertile soil due to high salinity or unsatisfactory crop yield. Over-irrigation, i.e., excessive water application, on the other hand, may lead to water-logging in soil for a long time or uneven moisture distribution. The result—incidences of crop rot, insects and pest manifestation, improper crop growth due to lack of essentials like oxygen and minerals, and so on.

Additional problems include water wastage, unproductive effort and great loss of money.

Sensors finding their way into irrigation

Strategically placed sensors and automated control system for irrigation

can make a great difference. The level of automation varies based on the user's requirement. The technology can range from simple timer-based controls to Internet of Things (IoT)-driven smart irrigation.

As regards application, sensors are used to read the moisture content and temperature of soil, as also weather conditions like humidity, rainfall and snow, and schedule the irrigation accordingly. Fertiligation is another major area of application for sensors and automation. Levels of residual fertilisers, pesticides and other chemicals in the soil are detected by the sensors, based on which components of fertilisers and other chemicals are calculated and scheduled for addition.

In electricity-deficient countries like India, automated systems can be applied to ensure sufficient amount of irrigation in a day, even if power availability is very poor. For example, if a one-hour-scheduled irrigation process is interrupted by a power cut within 30 minutes of action, the automated system will shut down by itself and resume the remaining 30 minutes of process as soon as electricity resumes.

For areas where regular on-field surveillance is difficult to arrange, automated irrigation takes up the charge. GSM-enabled devices help in scheduling as well as remote control of irrigation activities on the field. An IoT-powered setup sends irrigation data to the cloud, from where it can be used for further analytics.

Benefits

Substantial increase in crop yield and quality

This, of course, is the most vital benefit of automated irrigation. Optimised conditioning of soil and crops greatly influences the yield factor. While improvement in productivity varies, agri-owners can surely expect a yield increase between 15 and 30 per cent.

Swapnil S. Ingole, country manager, Galcon Smart, says, "Auto-